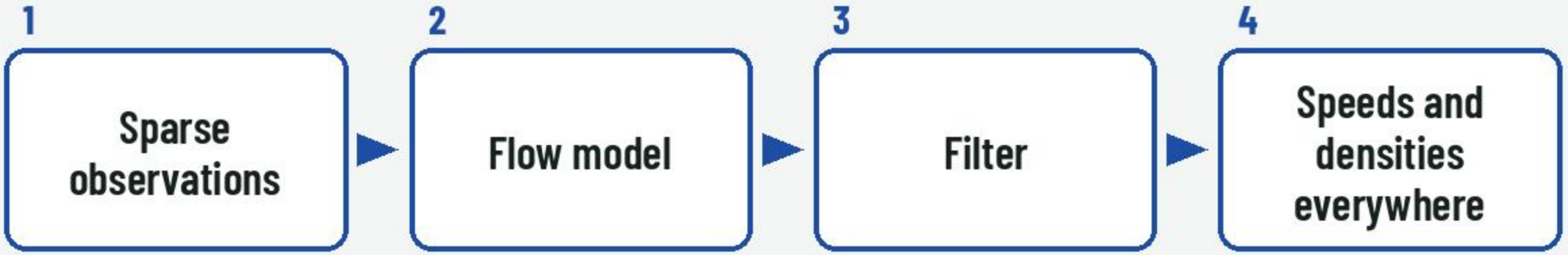
BUILDING THE TWIN

Estimation and prediction

Module 6 · From sparse data to a full picture

Transport Twin Study Path

Traffic state estimation



Two families of prediction

Model-based

Flow physics (LWR, cell transmission)

Explains itself

Needs calibration

Data-driven

LSTM, graph neural networks

Learns patterns

Needs lots of clean data

Hybrid and physics-informed

- Learn the residual a model gets wrong
- Constrain networks with vehicle conservation
- Less data, fewer impossible answers

Evaluate honestly

- MAE, RMSE, MAPE by horizon: 5, 15, 30, 60 min
- Beat simple baselines first
- Test on incidents and new months, not just normal days

Remember

A forecast is only useful in time

Prediction horizon must exceed decision and computation time